Architectural Particular Specification

JOINT SEALANTS

SECTION 6.5

Rev	Date	Description
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Section 6.5
JOINT SEALANTS

PART 1 GENERAL

1.01 SUMMARY OF WORK

A. Section Includes:

1. Preparing and priming sealant of substrate surfaces.
2. Provide and install high-performance elastomeric sealants, backings and required accessories.
3. Compatibility, adhesion, staining, and other specified sealant testings. Mock-ups and field testings.
4. Tooling of all sealant joints.
5. Compliance with the Sealant Manufacturer's Quality Assurance Programs and Procedures.
6. Furnishing and installing waterproof expansion joints in accordance with the details shown on the plans and the requirements of the specifications. The expansion joint shall be shall be a waterproof sealing system designed for split slab applications that utilizes heavy-duty aluminum frames with a continuous locking elastoprene seal that ties directly into the deck waterproofing.

B. Related Sections:

1. Section 2.1: CAST DECKS AND UNDERLAYMENT
2. Section 5.1: CARPENTRY, LAMINATE AND JOINERY
3. Section 6.1: DAMPPROOFING AND WATERPROOFING
4. Section 6.4: FIRESTOPPING
5. Section 8.10: GLAZING
6. Section 8.3: PAINTING
7. Section 8.9: SPECIAL COATINGS
8. Section 10.2: SANITARY FITTINGS AND ACCESSORIES

1.02 REFERENCE

A. Section Includes:

1. BS2499:1973 – Specification for Hot Applied Joint Sealants for Concrete pavements.
2. BS2499: Part 2: 1992 – Code of Practice for the Application and Use of Joint Sealants.
3. BS5212: Part 1: 1990 – Specification for Joint Sealants, Cold-Applied for Concrete Pavement.
4. BS5212: Part 2: 1990 – Code of Practice for the Application and Use of Joint Sealants, Cold Applied for Concrete Pavement.
5. BS5889: 1989 – Specification for One-Part Gun Grade Silicone - Based Sealants.
6. BS6093: 1993 – Code of Practice for Design of Joints and Jointing in Building Construction.
7. BS6213: 1982 (1992) – Guide to Selection of Constructional Sealants.
8. BS6262: 1982 – Code of Practice for Glazing for Buildings
9. BS EN 27389: 1991 – Building Construction Jointing Products.

10. BS EN 28340: 1991 – Building Construction Jointing Products. Determination of Tensile properties at Maintained Extension.
11. BS EN 28394; 1991 – Building Construction Jointing Products. Determination of Extrudability of One - Component Sealants.
12. BS EN 29046: 1991 – Building Construction Sealants. Determination of Adhesion/Cohesion Properties at Constant Temperature.

B. American Standard:

1. STM C510 – Test Method for Staining and Color Change of Single- or Multicomponent Joint Sealants.
2. STM C603 – Test Method for Extrusion Rate and Application Life of Elastomeric Sealants.
3. ASTM C639 – Test Method for Rheological (Flow) Properties of Elastomeric Sealant.
4. ASTM C661 – Test Method for Indentation Hardness of Elastomeric-Type Sealants by Mean of a Durometer.
5. ASTM C679 – Test Method for Tack-Free Time of Elastomeric-Sealants.
6. ASTM C717 – Definition of Terms Relating to Building Seals and Sealants.
7. ASTM C719 – Test Method for Adhesion and Cohesion of Elastomeric Joint Sealants Under Cyclic Movement (Hockman Cycle).
8. ASTM C792 – Test Method for Effects of Heat Aging on Weight Loss, Cracking, and Chalking of Elastomeric Sealants.
9. ASTM C793 – Test Method for Effects of Accelerated Weathering on Elastomeric Joint Sealants.
10. ASTM C794 – Test method for Adhesion-in-Peel of Elastomeric Joint Sealants.
11. ASTM C1021 – Laboratory Testing of Sealants.
12. ASTM C1193 – Guide for Use of Elastomeric Joint Sealants.
13. ASTM D2377 – Test Method for Tack-Free Time.
14. ASTM E1399 – Test method for Cyclic Movement Joint Width.

1.03 QUALITY ASSURANCE

Qualification of Applicator:

A. Company specializing in high-performance elastomeric joint Sealant application, with minimum of five years documented experience, certified and approved by the Sealant Manufacturers specified herein.

1.04 PERFORMANCE REQUIREMENTS

Sealant Performance and Design Requirements:

1. All joint sealants and structural glazing sealants used for this Project shall be high-performance elastomeric sealants with a minimum expected service life of 20 years.

2. Sealants used as weather seals shall not experience adhesive or cohesive failure. Sealants shall withstand movements up to the limits required by joint design calculations and Sealant Manufacturer's recommendation. Exposed sealant surface shall not crack or bubble. Sealants, primers, joint fillers and other joint/sealant accessories shall not stain adjacent materials. Sealants shall be used only if the sealant manufacturers' adhesion, compatibility stain and other tests, indicate in 1.11 below, yield favourable results. Sealants shall not be adhered to, or placed against, the edge of a laminated glass unit inter-layer.
3. Compatibility and other testings between each different sealant and with primers, joint fillers backer rods, bond breakers and other Sealant accessories shall be conducted to confirm suitability. The Management Contractor and Joint Sealant Sub-Contractor shall coordinate with other Sections to assure compatibility of intersecting sealants.
4. Sealant Depth:
   a. The recommended sealant width and depth relationships shall be determined by the manufacturer of the sealant. Generally accepted guidelines (which shall be confirmed by the Sealant Manufacturer) are as follows:
   b. For a recommended minimum sealant width of 6.0mm, the sealant depth should be 6.0mm. For joints in concrete, masonry, or stone, the depth of the sealant may be equal to the sealant width in joints up to 13mm. For joints from 13mm to 25mm wide, the sealant depth should be one half the width. For expansion and other joints in which the sealant is 25 to 50mm wide, the sealant depth shall not be greater than 13mm when the sealant is applied. For sealant width exceeding 50mm, the depth shall be determined by the Sealant Manufacturer.
   c. For sealant widths in metal, glass, and other nonporous surface joints, the sealant depth should be minimum of 6mm for widths of 6mm to 13mm, a minimum of one half the width up to 13mm, and in no case exceed 13mm.
   d. Where backer rods are used, depth of joints shall be measured to the crown of the round or semi-circular rods.

B. The Contractor shall strictly adhere to the Sealant Manufacturers' Quality Assurance Programs and Procedures to ensure the warranty of the Work of the Section.

C. The Contractor shall submit the Sealant Manufacturer's Quality Assurance Programs and Procedures (incorporating all the requirements of this Section.) to the Architect.

1.05 ENVIRONMENTAL REQUIREMENTS

A. Do not install solvent curing sealants in enclosed building spaces without adequate ventilation, conform with Manufacturer's recommendation

B. Maintain temperature and humidity recommended by Sealant Manufacturers during and after installation, and while in storage.

1.06 ENVIRONMENTAL CLIMATIC & PSYCHOMETRIC CONDITIONS

A. The Contractor and Sealant Manufacturer shall warrant that all products, materials and items are suitable, and weatherproof and elastomeric seal can be maintained in the various climatic conditions encountered.

1.07 SUBMITTALS

A. Propose sub-contractor specializing in installation of sealant system with ten (10) years minimum experience on projects of a similar size and complexity.

B. Where reference is made to products from various manufacturers, submit a list of the proposed products and confirm compatibility between all elements/materials as appropriate.

C. Alternatively products from a single source supplier are permitted. Confirm compatibility between all elements/materials as appropriate.

D. The Management Contractor shall submit four sets of wall cladding, window, and glazing shop drawings with joint sealant and structural glazing details and locations to each Joint Sealant Manufacturer for review and comment. Each Joint Sealant Manufacturer shall returned 1 set of mark-up, reviewed shop drawings to the Employer's Project Manager, Joint Sealant Applicator, the Main-Contractor, and the Architect.

E. Submit product and technical performance data, method statement for application, and samples as required in Section 01 33 00.

F. Sealant testing submittals as required in this Section.

G. Submit samples of all custom colours for the Architects review and approval.

J. General – Submit the following according to Division 1 Specification Section.

K. Standard Drawings – Submit typical expansion joint drawings indicating pertinent dimensions, general construction, blockout dimensions and product information.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY, STORAGE AND HANDLING

A. Materials shall be delivered in Sealant Joint Material Manufacturers' containers, dry, undamaged, unopened packages, all clearly labelled with the manufacturer's name, product identification, expiration date, pot/shelf-life, and lot numbers intact.

B. Materials shall be stored in strict accordance with the Manufacturer's printed instructions, copies of which will be furnished to the Employer's Construction Manager and the Architect.

C. All lots and/or shipment of joint sealants and accessories shall be tested within 30 days of receipt of materials according to the procedure for determining field sealant quality as stated in Clause below. Outdated materials may not be used. The Sealant Manufacturer's Representative shall verify that all lot and/or shipments are from an authorized distributor and the sealants and accessories are genuine products of the specified Sealant Manufacturers.

D. The special precautions recommended by the Manufacturer shall be strictly followed where flammable and hazardous materials are involved or released. Hazardous waste materials shall be legally transported and disposed of in strict conformance to local regulations.

E. Deliver products to site in Manufacturer's original, intact, labeled containers and store under cover in a dry location until installed. Store off the ground in temperatures above 40ºF, protect from weather and construction activities.

1.10 MOCK-UP

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTINGS

A. Test Methods

1. Standard Conditions for Laboratory Tests - All tests described below shall be performed in a laboratory controlled at 23 + 2°C temperature and 50 + 5% relative humidity. The sealant sample shall be conditioned at this temperature and relative humidity for at least 24 hours before laboratory tests are made for the following:

   a. Rheological Properties – Test Method ASTM C639

   b. Hardness – Test Method ASTM C661

   c. Application Life – Test Method C603

   d. Hardness – Test Method ASTM C661

   e. Tack-Free Time – Test Method ASTM C679

   f. Stain and Color Change – Test Method ASTM C510

   g. Adhesion and Cohesion After Cyclic Movement – Test Method ASTM C719

   h. Adhesion -in Peel – Test Method ASTM C794

   i. Glass – Test Method ASTM C794

   j. Effects of Accelerated Weathering – Test Method ASTM C793

2. Allow minimum of 8 weeks in the Programme of Work for sealant testings by the Joint Sealant Manufacturers. The Programme of Work shall indicate testing time allowances.

B. Submittals for Sealant Testings.

1. The Joint Sealant Sub-Contractor shall submit to the Sealant Manufacturers complete two sets of complete shop drawings identifying the Project and location, name and address of the Architect, dimensions of lites (w x l), sealant contact width and depth dimensions, specified design wind loads, joint locations and finish/substrates.

2. The Joint Sealant Sub-Contractor shall submit to the Sealant Manufacturers samples of all substrates and sealants of other manufacturer, which either come in contact with, or are in close proximity to the joint sealant and structural glazing sealants for adhesion, staining, structural glazing adhesion, and compatibility testing. All substrate and material samples shall have the applied finishes, coatings, etc. which will be installed in place on the Project. The following samples are required:

   a. Metals - nine 200mm x 200mm size long production samples of each type of metal with the type finish identified.

b. Glasses - Nine 200mm x 200mm minimum production lots of each type of glass with glass type, coating (if required) and manufacturer identified. Gaskets, Spacers and Setting Blocks - One 300mm x 300mm size sample of each with manufacturer and type identified.

c. Other material or finishes - Nine 300mm x 300mm size of each type of material with applied finish or coating.

3. Furnish and install all joint sealants, backings and accessories as required for all field mock-ups and field testings. Labour and material for the construction and testing of mock-ups and field testings shall be included in the Tender sum.

C. Test Reports

1. Submit a certified laboratory test report in accordance with BS 5368: Pt. 4. include the following for the tests carried out under Clause 16.73 :-

a. Test results together with comparison with permitted parameters.

b. Remarks and conclusions by the testing professionals.

2. Submit photographic records of the test arrangement, set-up and of the performance of the mock-ups, and constituent components of the same, during fabrication, assembly and under test. Submit dated photographic records one set each to the Employer's Construction Manager, the Architect, and the Sealant Manufacturers.

3. Provide four copies of the agreed shop drawings for recording any modifications found necessary during and after the laboratory tests. The testing laboratory shall accurately and neatly record all changes, revisions, modification etc., made to the shop drawings. At the completion of the testing, one copy each of the marked-up drawings shall be given to the Management Contractor, the Joint Sealant Sub-Contract and the Architect.

4. Subsequently, and before any installation Work commences on the Project Site, the Joint Sealant Sub-Contractor shall provide the Employer's Construction Manager and the Architect with on reproducible and two prints of the approved shop drawings incorporating the modifications made.

D. Sealant Tests.

1. Field Adhesion Tests

a. All lots and/or shipments of joint sealants shall be tested within 30 days of receipt according to the procedure for determining field sealant quality using a field adhesion test using methods recommended by sealant manufacturer.

b. Results of all tests shall be recorded in a Sealant Field Test Log Book provided by the Joint Sealant Manufacturer and kept on the Project Site by the Management Contractor. It shall be available for review upon request by the Employer's Project Manager, the Architect or the Joint Sealant Manufacturer. The Log Book shall be submitted as part of the Project Close-Out Submittal. The Log Book shall be protected by a waterproof covering and prevented from damage.

c. All sealants held longer than five months after initial testing must be retested to determine field quality before use. The Sealant Manufacturer's Technical Field Representative shall confirm that all sealants retested can be used prior to installation. All retesting shall be recorded in the Sealant Field Test Log Book. No sealants that have started to set in their container or have exceeded their shelf/pot life shall be used.

2. Sealant Physical Requirement Tests

The contractor is to submit evidence that the proposed sealants meet or exceed the following performance requirements.

a. Rheological Properties:

1) Grade P (pourable or selfleveling) sealant shall have flow characteristics such that when tested in accordance with Test Method ASTM C 639 it shall exhibit a smooth, level surface. (Refer to Types I and III in the test.)

2) Grade NS (nonsag) or gunnable sealant shall have flow characteristics such that when tested in accordance with Test Method ASTM C639 it does not sag more than 4.8mm in vertical displacement. Also the sealant shall show no deformation in horizontal displacement. (Refer to Types II and IV in the test.)

b. Extrusion Rate:

Type S (single component). Grade P (pourable or selfleveling) sealant shall have an extrusion time of not more than 20 seconds when tested in accordance with Test Method ASTM C603.

i) Type S (single component), Grade NS (nonsag or gunnable sealant) shall have an extrusion time of not more than 45 seconds when tested in accordance with Test Method ASTM C603.

c. Application Life:

i) Type M (multicomponent), Grade P (pourable or selfleveling) sealant, when tested in accordance with Test Method ASTM C603 must require no more than 20 seconds to empty the cartridge 3 hours after mixing.

ii) Type M (multicomponent), Grade NS (nonsag or gunnable) sealant when test in accordance with/Test Method ASTM C603 must require no more than 45 seconds to empty the cartridge 3 hours after mixing.

d. Hardness:

i) Use T (traffic) sealant shall have a hardness reading after being properly cured, of not less than 35 seconds or more than 50 when tested in accordance with Test Method ASTM C661.

ii) Use NT (nontraffic) sealant shall have a hardness reading, after being properly cured, of not less than 15 or more than 50 when tested in accordance with Test Method ASTM C661.

e. Effects of Heat Aging: The sealant shall not lose more than 10% of its original weight or show any cracking or chalking when tested in accordance with Test Method ASTM C792.

f. Tack-Free Time: There shall be no transfer of the sealant to the polyethylene film when tested at 72 hours in accordance with Test Method ASTM C679.

g. Stain and Color Change: The sealant shall not cause any visible stain on the top surface of a white cement mortar base when tested in accordance with Test Method ASTM C510.

h. Adhesion and Cohesion Under Cyclic Movement: The total loss in bond and cohesion areas among the three specimens tested for each surface shall be no more than 9 cm^2 when test in accordance with Test Method ASTM C719 with standard mortar, glass, and aluminum or any other specified substrates.

i. Adhesion-in-Peel: The peel strength for each individual test shall not be less than 2.3kg when tested in accordance with Test Method ASTM C794 with standard mortar, glass, and aluminum or any other specified substrate. In addition, the sealant shall show no more than 25% bond loss for each individual test.

j. Adhesion-in-Peel for Use G Exposed to Ultraviolet Exposure Through Glass: The peel strength for each individual test shall not have less than 2.3kg and the compound shall be no more than 25% bond loss for each individual test when tested in accordance with Test Method ASTM C794.

k. Note: Curing conditions are specified by all of the test method cited. The manufacturer may request other conditions than those specified for the curing period provided they meet the following requirements: (1) the curing period shall extend for 21 days; (2) the temperature during the curing period shall not exceed 70°C; and (3) the amended curing conditions recommended by the manufacturer shall also be applied to the durability, adhesion in peel and ultraviolet radiation exposure tests.

l. Effects of Accelerated Weathering: The sealant shall show no cracks greater than those shown in Example #2 of Fig. 1 of ASTM C962 after the specified ultraviolet exposure and shall show no cracks greater than those shown in Example #2 of Fig. 2 of ASTM C962 after exposure at cold temperature and the bend test when tested in accordance with Test Method ASTM C793.

1.12 WARRANTY

A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

B. Provide Manufacturer’s ten (10) year warranty for all joint sealants and accessories against material defects and workmanship.

C. Nothing shall be done by the Contractor which will void any Manufacturer’s Warranty.

D. Warranty:

1. The applicator and the manufacturer shall jointly provide guarantee for the performance of the sealants and accessories with a form approved by architect, warranty all works in connection with the specified system to be free from defects in material and workmanship for a period of 10 years from the date when the relevant works is completed.
2. Include coverage for installed sealants and accessories which fail to achieve airtight seal, exhibit loss of adhesion or cohesion, or do not cure.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURERS

A. Subject to compliance with the Project Contract Documents requirements, provide joint sealants and required sealant accessories and joint fillers from one or more of the following manufacturers or equal approved by the Architect:

1. Tech Link Construction Engineering Limited
   Unit 802, Laford Centre
   838 Lai Chi Kok Road
   Kowloon
   Tel: (852) 3701 1817 / 9316 8906 Fax: (852) 3701 1818

   Email: kenuskwong@techlinkcon.com.hk

2. or approved equivalent

2.02 MATERIALS

A. Provide and install joint sealant, or equivalent approved by the Architect, as listed below, and as shown on the Drawings, with ancillary sealant materials necessary for a proper, complete, weather-tight, fire-resistance requirement and elastomeric sealant installation. (This is a partial list of principal items only).

B. Refer to the Contract Documents for items not specifically scheduled. Refer to Section 8.10 Glazing for glazing sealants.

C. Sealant Type

The sealant type listed below are subject to test verification and recommendation made by the respective Joint Sealant Manufacturers listed herein.

1. Tile Joint Sealant (For Water Tank): (refers to section 09 30 00)
   a. Approved Products: "PCI Silcoferm KTW" silicone joint sealant in drinking water applications or equivalent complying with KTW and DVGW worksheet W270 requirements.

2. Tile Joint Sealant (For Flat Roof, Podium Deck, All internal wet area e.g. Lavatory / Plant room / Pump room / Refuse room and external wall with rendering finishes, etc.)
   a. Approved Products: "PCI Silcofug E" silicone joint sealant discouraging fungal decay and mildew growth or equivalent

3. Closed Cell Back Rod;
   a. Approved Products: Tech Link backer rod or approved equivalent. It shall be round, flexible, continuous lengths of extrude, closed-cell polyethylene foam. Available in variety of diameters (from 6-70mm)

4. Proprietary Expansion Joint Cover;
   a. Approved Systems: MM HFX Systems (includes EHFXF) or approved equivalents consisting of aluminum frame, slide, cover plate along with appropriate fasteners and anchors.
   b. Minimum Performance :
      i) Aluminum conforms to ASTM B221 for extrusion alloys 6063-T5 and 6063-T5 and 6061-T6
      ii) Aluminum conforms to ASTM B209 for sheet and plate

iii) Steel conforms to ASTM A569 for commercial quality sheet and plate

iv) 204-R1 conforms to AA-M10C22A31

v) Excluded Preformed Seals PVC Profiles conform to ASTM C920

vi) Conforms to ASTM E1399 for cycling joints

5. Non-fire Rated Joint Filler

a. Approved Products: Tech Link joint filler or approved equivalent. It shall be available in verity of thickness (from 10 – 50mm) and size in 1m x 2m/ pcs.

D. Sealant Colours

1. All exterior sealants, sealants within interior lobbies, toilets shall be custom colours to match adjacent colours of materials, finishes, or as indicated on the Drawings. All custom colour samples shall be reviewed by the Architect for approval.

2. All other sealants shall be of Manufacturers' standard colours selected by the Architect.

3. All custom and standard colour sealants shall be warranteed against discoloration, chalking, ultraviolent and ozone degradations, adhesion and cohesion failures, and other material defects.

2.03 ANCILLARY MATERIALS

A. General: All Sealant accessories used with any sealant or substrate shall be compatible with each other. Where unanticipated condition arise in the field that may be detrimental to good sealant, primer or other sealant accessory performance, field testing shall be conducted by the Sealant Manufacturer's Technical Representative to determine proper treatment, and sealant and accessory selection.

B. Primers: Non-staining type as recommended by the Sealant Manufacturer to suit each application and each substrate type.

C. Joint Cleaners: Non-corrosive and non-staining type as recommended by the Sealant Manufacturer and compatible with each type of joint forming materials and accessories.

D. Joint Fillers:

1. Non-corrosive and non-staining type as recommended by the Sealant Manufacturer and compatible with joint forming materials and accessories Joint filler shall be oversized by 30% to 50%. Joint filler material shall not be impregnated with oil, bitumen, non-curing polymers or similar materials. Joint filler shall not adhere to Sealant. And adhesive backed, bond breaker tape must be installed to prevent three-sided adhesion.

2. Joint fillers are classified as Type A or Type B in accordance with ASTM C962:

a. Type A, those whose main purpose is to control the depth of the sealant in the joint and to permit full wetting of the intended interface when tooled.

b. Type B, those which serve the same functions as Type A, but, in addition, can serve as a temporary joint seal for weather protection or as a secondary barrier behind the prime sealant barrier, or both.

3. Type A Joint-Fillers-Materials used for Type A joint-filler application may be flexible and compatible closed-cell plastic foam or sponge rubber rod stock. They should be capable of resisting permanent deformation before and during sealant application, should be non-absorbent to water and gas and not blow upon mild heating as this can cause bubbling of the sealant. Caution shall be exercised when installing the joint-filler to avoid puncturing or over compression which might also lead to bubbling. A sealant manufacturer may recommend or approve open-cell sponge type materials such as urethane foam. These may be satisfactory, provided their water-absorption characteristics are recognized. They shall not be employed where possible absorption of water may be detrimental to sealant performance. The sealant shall be applied immediately after joint-filler placement to prevent water absorption from rain. Due to the ease of compressibility of open-cell foam joint-fillers, special sealant tooling techniques are required.

4. Type B Joint-Fillers-Type B joint-fillers may be compatible elastomeric tubing of such materials as neoprene, butyl, or EPDM. These may be applied immediately as a temporary seal until the primary sealant is applied, after which they serve to a limited degree as a secondary water barrier. They shall be non-absorbent to water and gas and not blow upon mild heating or rupturing, as this can cause bubbling of the sealant. In addition, these joint-fillers shall have the ability to remain resilient at temperatures as low as -26°C and have low compression set.

5. Horizontal Deck Joint Application - Joint-fillers used for floors, pavement, and other light traffic area may be compatible extruded closed-cell high-density flexible foams, cork board, resin-impregnated fibreboard or elastomeric tubing or rods. These joint-fillers shall remain resilient to as low as -26°C, exhibit good recovery, shall not cause the sealant to bubble in the joint due to heat or rupturing, and shall be non-absorbent too water. In addition, they shall be capable of supporting the sealant in traffic areas. Joint fillers shall not exude liquids under compression, as this could hydraulically cause sealant failure by forcing the sealant from the joint. In horizontal joints over 50mm in wide, install Type 316 stainless steel with 2.5% molybdenum, 1.5mm thick minimum metal plates underneath a bond breaker used under the sealant to support it. (Figure 2 of ASTM C962.)

6. When the time arrives to install the sealant, the recess above the joint-filler may be wet and contaminated or may not be compressed against the substrates due to shrinkage of concrete or thermal expansion of the joint, permitting loss of sealant if a self-leveling type sealant is used. The problem may be resolved by using a combination of joint filler materials. Form a joint in concrete by installing a premolded joint filler, thus allowing a greater recess above and installing an additional joint-filler material under compression across the width and to the proper depth just prior to application of the sealant.

7. Joint Filler Schedule:

   a. Bitumen Impregnated Fibreboard Filler:

      Compressible, non-extruding, bitumen impregnated suitable for forming joints between in-situ and pre-cast components, structural expansion and separation joints in concrete highways pavements and floor. Conforms to AASHO standard spec. M153-54.

   b. Cross-linked, Non-Absorbent, Polyethylene Joint Filler:

      For forming expansion joints in concrete, brickwork and block work. Recommended for water retaining or excluding structures subjected to hydrostatic pressure. Suitable for use in contact with potable water (BS6920). Nominal density of 60kg/m³.

E. Bond Breakers: Polyethylene tape with pressure-sensitive adhesive on one side, or other bond breakers as recommended by the Sealant Manufacturer. Install bond breaker to the bottom of joints containing rigid, non-flexible joint filler materials or mortar to preclude the possibility of the sealant adhering to these materials and forming three-sided adhesion. Materials impregnated with oil, bitumen, non-curing polymers, and similar materials shall not be used as bond breakers.

F. Backer Rods: Refer to Section 8.10 Glazing.

PART 3 EXECUTION

3.01 EXAMINATION/INSPECTION

A. Verify that surfaces and joint openings are ready to receive Work and field measurements are as shown on approved Shop Drawings and recommended by the Sealant Manufacturer.

B. Beginning of installation means the Contractor's acceptance of existing substrates and conditions.

3.02 PREPARATION

A. All surfaces to receive sealants and accessories shall be clean, dry, free of loose materials, dirt, dust, rust oil, and other contaminants.

B. Concrete and masonry surfaces shall be cured, then cleaned, by manual or power brushing or grinding, blast cleaning with oil-free compressed air or vacuumed to remove dust of cleaning. Solvent shall not be used on concrete or other porous surfaces. Concrete surfaces must be free of release agents, curing compounds or other adhesion inhibiting contaminants.

C. All non-porous surfaces which will be in contact with sealants shall be cleaned with an approved solvent. Small areas shall be washed and then dried with clean cloth before the solvent evaporates. This final cleaning shall be done after other necessary preparations have been completed.

D. Cleaning of Substrates:

1. Substrates like glass and aluminum are not porous and must be cleaned with a solvent before the sealant is applied. The solvent selected depends upon the substrate and the type of dirt or oil that is required to be removed. Always check with the supplier to assure that the cleaning procedure and solvents are compatible with each substrate. Non-oily dirt and dust can generally be removed with a 50% solution of Isopropanol and water. Oily dirt or films generally require a degreasing solvent such as MEK or Xylene.

2. Concrete, granite, marble, and limestone are porous, and depending on the surface condition, may require abrasion cleaning, solvent cleaning or both. Abrasion cleaning can be done by grinding, saw cutting, sand and water blasting, mechanical abrading or a combination of these methods. Remaining dust or loose particles should be removed by vacuuming or by application of high pressure oil free, compressed air. If there is still dirt on the surface, then it must be solvent cleaned. Some porous surfaces will trap solvents after cleaning or priming. This solvent must be allowed to evaporate before sealant is applied.

3. The Two Cloth Cleaning Method must be used whenever solvents are used for cleaning. Clean, soft absorbent, lint-free, dye-free cotton cloths shall be used. The method consists of a solvent cloth wipe followed by a dry clean cloth wipe. Pour or squirt the solvent onto the cloth, a plastic squeeze bottle works well. Never dip the cloth into a container of solvent as this will contaminate the bulk solvent. Wipe vigorously to remove surface contaminants. Check the cloth to see if it has picked up contaminants. Rotate the cloth to a clean area and rewipe until no more dirt or oil material is picked up. Immediately wipe the solvent cleaned area with a clean, dry cloth. The solvent must be removed with a dry cloth before the solvent evaporates, otherwise the cleaning will be ineffective. Primer, if necessary, can then be applied to the clean dry surface. Some solvents are hazardous and may require special attention to ventilation or they may have environmental or handling restrictions. Consult the solvent supplier for recommended safe handling and disposal procedures.

4. Satisfactory adhesion only occurs on a clean surface. The work area must be clear of airborne dust during the cleaning period. Fingerprints are very difficult to remove from metalised surfaces and care must be taken when handling substrates. Also ensure surface condensation does not occur after cleaning.

E. Priming:

1. Primers are moisture sensitive materials and will be de-activated if contaminated by water or humidity while in the can. Expired, milky primer or primer containing a white precipitate shall be discarded.

2. Never dip into a container of primer. Always pour from the container to avoid contamination. The preferred technique for primer application is with a cloth. It is recommended that either cotton wool or a 100% cotton cloth which is clean, dry and lint-free and dye-free be used.

3. A thin film of primer shall be applied. This is the most critical step. If a white powdery residue appears, too much primer has been applied. Excess primer can be removed prior to sealant application by simply wiping off with a clean cotton cloth. Primer accidentally splashed onto metal or glass can be best removed by washing with iso-propyl alcohol or toluene.

4. When the prime coat is dry to touch, sealant application can begin. This is generally 30 minutes.

5. Primer shall only be applied to areas where the sealant will be applied upon it within the same working day. Priming too far in advance of sealant application can result in contamination of the primed surface due to dust, oil or water which may render the primer ineffective.

F. Masking:

1. Mask all joints prior to the application of sealant.

2. Mask the exterior face adjacent to the joint with tape runs of approximately 900 to 1800 mm in length, starting from the top down and overlap the runs.

3. Masking shall be removed immediately after tooling of sealant.

3.03 INSTALLATION/APPLICATION

A. Backing Application:

1. Backer rod shall be of proper size to provide a thickness of sealant in the joint to comply with Sealant Manufacturer’s recommendations Polyethylene backer rods (if specified) shall not be punctured during installation. Sealant bubbling may result from outgassing of the backer rod.

2. Joint backing shall be thoroughly dry. Do not install more joint backing (and bond breaker if used) than can be sealed in one day. Install joint backing to achieve accurate depth of joint as recommended by the Sealant Manufacturer for each application, each substrate and each sealant type.

3. Polyurethane foam joint backing is not approved for horizontal deck joint or fire-resistive joint application.

B. Sealant Application:

1. Application of the sealant may be by cartridge type caulk gun or bulk type, air pressure activated. To the properly prepared joint, apply the sealant as a full bead starting from the bottom of vertical joint and working up. Push the bead ahead of the air nozzle to insure complete filling of the joint. Air pockets, foreign embedded matter, ridges, sags, or voids along sealant bead are not acceptable.

2. Where triangular fillet sealant application is necessary, it shall be applied so that fillet is not less than 10mm across the face and adhere a minimum bearing of 6mm on both faces. Where there is a gap between adjoining faces greater than 5mm, a back-up material shall be inserted. Finish seal shall have a rounded (convex) surface and not concave - so that there is sufficient body of sealant. All triangular fillet sealant application must have prior approval by the Sealant Manufacturers.

3. Time of Sealant Application:

a. Ideally, the best time for application of sealant to movement joints is when the joint gap width is at the mean, tending to the maximum with elastic sealants. This reduces tensile strain.

b. Application of sealant at high temperature will reduce sealant viscosity and may result in slump or flow especially in wide joints. Working life may also be unacceptably reduced. Consult Joint Sealant Manufacturer for recommendation.

c. Application of sealant at low temperature can increase sealant viscosity, giving rise to mixing (in the case of 2-part sealants) and application difficulties, and reduction in wetting of component interfaces. Sealant cure will be reduced, or even arrested, at low temperature. Consult Joint Sealant Manufacturer for recommendation.

d. High moisture content of substrates are detrimental to sealant adhesion. Sealing shall not be undertaken if free moisture or frost is present on the joint faces.

4. Tooling:

a. Tool sealant with light pressure to spread the material against the back-up material and the joint surfaces. Tool immediately after sealant application and before a skin forms.

b. Do not use soap or oil as a tooling aid. Remove masking as soon as each sealant bead is tooled.

c. Uncured sealant may be cleaned from tools and non-porous surfaces using commercial solvents such as xylol. On porous surfaces allow excess sealant to cure and then remove it by abrasion or other mechanical means. Re-prime joint as required.

d. A clean tool with a concave profile shall be used for all tooling work.

5. Cure time is dependent on the depth of the sealant joint and relative humidity. With structural glazing, temporary blocking of stops must be used to hold the glass immobile until the sealant has cured for approximately 14 to 21 days at temperatures above 10°C. Removal of temporary blocking or stops before sealant is totally cured will apply stresses on the partially cured sealant, which will affect its adhesive or cohesive properties.

6. All sealants are to be test verified for compatibility with the following conditions or substrates. Do not apply sealant in the following conditions or substrate unless it is specifically tested and approved by the Joint Sealant Manufacturers :

   a. To surfaces, tanks or pool in continuous water immersion, nor for below ground joints or where excessive abrasion and physical abuse are encountered.

   b. To building materials that bleed oils, plasticizers, or solvents materials such as impregnated wood, oil-based caulks, and certain green or partially vulcanized rubber gaskets and tapes.

   c. In totally confined spaces where sealant is not exposed to atmospheric moisture.

   d. To surfaces which will be painted. The paint film will not stretch with ‘the extension of the sealant and may crack and peel

   e. To surfaces in direct contact with food, potable water, salt water, aerobic and anaerobic bacteria, other adverse conditions, or unfavorable substrate.

7. Delayed Assembly

   a. The recommended practice for the application of sealant is to assemble all substrates in their final position, and then applies the sealant into the joint.

   b. Application techniques where substrates are brought in contact with already applied sealant, is defined as delayed assembly. This practice of delayed assembly is not permitted because of the risk of the sealant curing prior to the time that the second substrate is brought and locked into place.

8. Apply Sealant and accessories to recommended substrate conditions and within recommended, temperature and humidity ranges. Consult with the Sealant Manufacturers when sealant and accessories cannot be applied within recommended parameters.

9. Aluminum mill finish is not acceptable at structural silicone bonding surfaces.

10. Aluminum surface to which structural silicone will be adhered shall have a finish which demonstrates by test its compatibility with specified joint sealant requirements.

C. Sealer Application [JS01] BASF PCI Naturstein-Impragnierung:

1. Apply BASF PCI Naturstein-Impragnierung undiluted with a roller or a paint brush in a very thin layer to cover all surfaces of the stone, where the stones to be installed on floor only (For stone to be installed on wall, apply the BASF PCI Naturstein-Impragnierung on the top surface of the stone only)
2. A repeated wet-on-wet application increases the effectiveness. Only apply as much of the impregnation as can be absorbed by the substrate.
3. Allow the product to react for approx. 1 hour, then remove any surplus with a dry, absorbent cloth. Afterwards remove the residues by polishing with a clean dry cloth not leaving any streaks.
4. PCI Naturstein-Impragnierung takes about 3 to 4 hours to dry; it should be avoided during this time to walk on and touch the surface or expose the surface to moisture. The impregnation develops its full effectiveness after approx. 24 hours.
5. Spray application is not allowed. Inhalation of atomized spray/ aerosol os harmful to health.

3.04 CLEANING

A. The surfaces of materials adjacent to the joints where sealant was applied shall be cleaned free of excess sealant or other soiling due to sealing operations. Excess sealant shall be carefully scraped (without scratching substrate) from the surface, and the remainder should be cleaned with xylol ormineral spirits. The surfaces shall be cleaned as work progresses, and before sealant begins to cure.

3.05 PROTECTION

A. All work adjacent to sealant shall be masked prior to the application of sealant.

B. Drop cloths shall be provided over all horizontal surfaces liable to receive dropping of sealant operations.

C. Misapplied sealants and droppings shall be immediately removed by methods and materials recommended in writing from the Sealant Manufacturers.

D. After material is applied and tooled, the Contractor shall remove all masking and other protection, and clean up any remaining defacement caused by his Work.

END OF SECTION